molecular

Type-I

→ Vibrational energy of the molecule in the upper electronic state exceeds the maximum value

After absorption

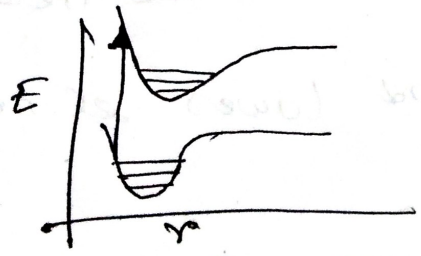
→ ↓ 1st oscillation dissociation

→ lower vibrational levels in the lower electronic state will not dissociation

→ In the continuous region dissociation

→ In discrete or banded structure ~~dis~~ not dissociation

→ It will in electronically excited state



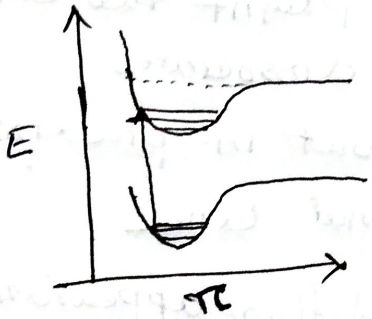
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Type-II

→ No transition lower to upper occur dissociation

→ Electronic spectrum consist bands with no continuous region

→ After absorption molecule only excited

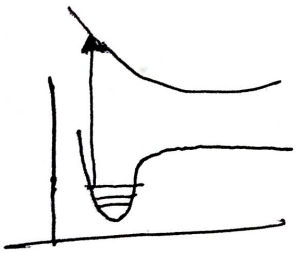


Type -III (Uncommon)

→ Upper electronic is unstable and has no vibrational levels

→ no bands but is continuous throughout

→ Electronic transition always by dissociation of molecules



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Type-IV (Predissociation)

→ combined II + III

→ Transition lower state to upper stable state

→ Two upper electronic state one is stable (II) and other unstable (III)

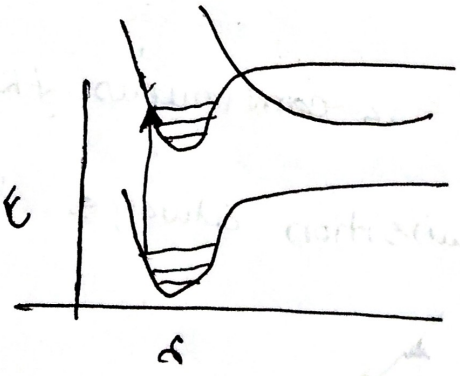
→ During vibration the excited molecules is changed into unstable state at point two curves cross one another then dissociates

→ Has banded structure but in predissociation level has absent rotational line

→ vibrational band having diffuse appearance.

→ Shorter period dissociation

→



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